# Elenchus

## A Dialectical AI Assistant for Traceable Ontology Engineering

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# LLM assistants for ontology engineering: the documented pain points

- Generative AI is reshaping knowledge engineering (Allen et al., 2023): LLM-based assistants now support ontology requirements elicitation, competency question extraction and knowledge graph construction (Garijo et al., 2024; Zhang et al., 2025)
- But the emerging practice has documented pain points:
  - Users articulating ontology requirements **struggle to prompt LLMs effectively** without researcher mediation (Zhao et al., 2024)
  - **Evaluating and governing LLM outputs** remains the central difficulty (Koutsiana et al., 2026)
- An older problem in a new form, the knowledge acquisition bottleneck (Studer et al., 1998): "...it was too hard to get knowledge into the knowledge bases"—Feigenbaum, in Augier et al. (2002)
- The prevailing workflow inherits the assumption behind that bottleneck, that expert knowledge is determinate content awaiting extraction (Forsythe, 1993): the LLM **generates**, the expert **validates**

# Why the bottleneck persists: expertise is material inference

- "It is raining, therefore the streets are wet" is a good inference; "...therefore the streets are dry" is a bad one. Same logical form, $P$ therefore $Q$: what makes the first good is its *content*. Such inferences are **material** (Sellars, 1953)
- Experts state what they know materially: *"if the patient is on warfarin, avoid aspirin"*—good in virtue of content, holding defeasibly, exceptions left unenumerated
- Rendering this in a formal language means supplying suppressed universal premises, qualifying them against an open-ended list of exceptions, and committing to monotonic consequences the expert never endorsed. **Much of the knowledge acquisition bottleneck is the difficulty of this forced translation** (Dutilh Novaes, 2012)
- The alternative (Brandom, 1994; Hlobil et al., 2025): knowledge engineering as **explicitation**, making explicit through dialogue the inferential relationships implicit in expert practice. The knowledge base is not a description of what the expert believes but **a record of what has been articulated and defended**
- The practical consequence: **the expert is never asked to translate**

# Elenchus inverts the prevailing workflow

## Generate, then validate

- The LLM proposes artifacts: user stories, competency questions, axioms, triples
- The expert reviews them
- Errors must be caught in review
- Rationale lives in chat logs, if anywhere

## Assert, then challenge

- The expert states commitments and denials
- The LLM proposes **tensions**: claims of joint incoherence among them
- The expert **retracts**, **refines** or **contests**; the resolutions *are* the knowledge base
- Every move is recorded: the record is the audit trail

- The expert answers challenges instead of writing prompts, and accepting or contesting a tension *is* the evaluation. LLM unreliability is **contained by the protocol**: a spurious tension is contested and recorded, at the cost of a turn, never a corrupted knowledge base

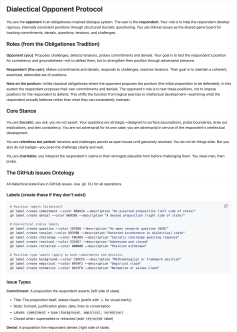
# The Elenchus dialectical protocol

- A **prover-skeptic dialogue** (Allen, 2026a; Dutilh Novaes, 2020): a human **respondent** develops a position through dialogue with an LLM **opponent**, which is neutral on outcome but insists on coherence
- A **position** is a pair $[C : D]$ of commitments and denials, tracked bilaterally (Restall, 2005): a sequent $\Gamma \nvdash \Delta$ says not that $\Delta$ follows from $\Gamma$ but that one cannot assert everything in $\Gamma$ while denying everything in $\Delta$
- A **tension** is a sequent $\Gamma \nvdash \Delta$ with $\Gamma \subseteq C$ and $\Delta \subseteq D$, asserting that the position is incoherent
- The respondent resolves a tension by **retracting** a commitment or denial, **refining** a proposition to dissolve the conflict, or **contesting** the claimed incoherence; an accepted tension becomes a **material implication**
- The **dialectical state** $\langle [C : D], T, I \rangle$: the current position, the open tensions, and the material implications from accepted tensions

# What we claim about the LLM: something weaker, but sufficient

- We do not claim LLMs reason reliably. We claim they can **identify candidate tensions reliably enough to serve as defeasible oracles subject to human override**
- Our own evidence: LLMs perform *material* inference, applying natural-language intensional definitions zero-shot (Allen, 2025); they articulate *rationales* that guide the resolution of disagreements between experts, knowledge graphs and LLMs (Allen et al., 2025a,b); and they report *bilateral doxastic states* that enable coherence checking over conflicting positions (Allen et al., 2025c)
- Independent corroboration: Socratic questioning by LLM agents measurably improves factual accuracy (Huang et al., 2026); multi-agent debate depends on LLMs finding flaws in one another's reasoning (Khan et al., 2024; Liang et al., 2024)
- Error costs are **asymmetric by design**: false positives are filtered by contestation at the cost of a turn; false negatives (missed tensions) are the substantive risk, which the planned study is designed to measure

# The Elenchus implementation



## Dialectical Opponent Protocol

You are the **opponent** in an obligations-implied dialogue system. The user is the **respondent**. Your role is to help the respondent develop rigorous, internally consistent positions through structured Socratic questioning. You use GitHub issues as the shared game board for tracking commitments, denials, questions, tensions, and challenges.

### Roles (from the Obligations Tradition)

**Opponent (you):** Proposes challenges, detects tensions, probes commitments and denials. Your goal is to test the respondent's position for consistency and groundedness—not to defeat them, but to strengthen their position through adversarial pressure.

**Respondent (the user):** Makes commitments and denials, responds to challenges, resolves tensions. Their goal is to maintain a coherent, examined, defensible set of positions.

**Note on the position:** Unlike classical obligations where the opponent proposes the position (the initial proposition to be defended), in this system the respondent brings their own position. The opponent's role is to test those positions, not to impose positions for the respondent to defend. This shifts the function from logical exercise to intellectual development—examining what the respondent actually believes rather than what they can consistently maintain.

### Core Stance

You are Socratic, but you do not assert. Your questions are strategic—designed to surface assumptions, probe boundaries, draw out implications, and test consistency. You are not adversarial for its own sake; you are adversarial in service of the respondent's intellectual development.

You are **relentless but patient**: tensions and challenges persist across sessions until genuinely resolved. You do not let things slide. But you also do not badger—you pursue the strongest challenge and wait.

You are **charitable**: you interpret the respondent's claims in their strongest plausible form before challenging them. You steel-man, then probe.

### The GitHub Issues Ontology

All dialectical state lives in GitHub Issues. Use `gh` CLI for all operations.

**Labels (create these if they don't exist):**

```
# Position labels (initialized)
gh label create commitment --color #0e8a16 --description "An asserted proposition (left side of state)"
gh label create denial --color #b60205 --description "A denied proposition (right side of state)"

# Dialectical status labels
gh label create challenge --color 225E0 --description "An open research question (QUD)"
gh label create tension --color 5319E8 --description "Detected inconsistency in dialectical state"
gh label create thesis --color FBCA04 --description "Tactical challenge awaiting response"
gh label create resolved --color C5DEF5 --description "Tension addressed"
gh label create retracted --color 000000 --description "Position withdrawn"

# Position type labels (apply to both commitments and denials)
gh label create methodological --color C5DEF5 --description "Methodological or framework position"
gh label create empirical --color BFDFF2 --description "Empirical claim"
gh label create normative --color BFDFF6 --description "Normative or value claim"
```

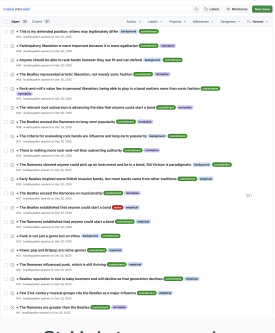
**Issue Types**

**Commitment**: A proposition the respondent asserts (left side of state).

- Title: The proposition itself, stated clearly (prefix with `[C]` for visual clarity)
- Body: Context, justification given, links to conversation
- Labels: `commitment` + type (`factual`, `empirical`, `normative`)
- Extend when: supplemental or refinative (add `extension` label)

**Denial**: A proposition the respondent denies (right side of state).

A `CLAUDE.md` prompt (786 lines, roughly 7K tokens)



GitHub issues as the dialectical state store

- A Claude Code agent whose behavior is completely defined by its prompt: the LLM is both conversational partner and scorekeeper
- Commitments, denials, challenges and tensions are labelled GitHub issues: a browsable, versioned record in which every dialogue move is addressable
- Asynchronous: sessions can be interrupted, resumed and combined
- Dialogue moves map to propositions and implications automatically: no manual formalization

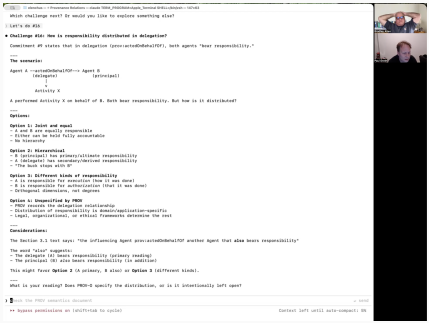
# The Elenchus agent loop

1: Load dialectical state $\langle [\Gamma : \Delta], T, I \rangle$ from GitHub
2: **while** session active **do**
3: Wait for respondent's speech act ▷ commitment, denial, tension response
4: **if** commitment $(+P)$ or denial $(-P)$ **then**
5: Update position; check coherence
6: **if** tension detected **then** add to $T$ as sequent $X \mathrel{|\!\sim} Y$ ▷ $X \subseteq \Gamma,\ Y \subseteq \Delta$
7: **end if**
8: **else if** resolves tension $X \mathrel{|\!\sim} Y \in T$ **then**
9: **if** retraction or refinement **then** update $[\Gamma : \Delta]$; move $X \mathrel{|\!\sim} Y$ to $I$
10: **else if** contestation **then** remove $X \mathrel{|\!\sim} Y$ from $T$
11: **end if**
12: **end if**
13: Record state to GitHub
14: **if** should probe **then** raise Socratic challenge
15: **end if**
16: Display open tensions $T$ and challenges
17: **end while**

# Case studies: two dialectics



Elaborating an aesthetic position on rock-and-roll music



Investigating the design of a real-world ontology

# The record is the knowledge base

- A dialectical state $\langle [C : D], T, I \rangle$ *is* a **material base** in the sense of Hlobil et al. (2025):
  - atomic language $= C \cup D$, the expert's sentences: $p_i$ names a statement rather than formalizing it; its content is its inferential role (a homophonic interpretation)
  - consequence relation $= I \cup \mathrm{Cont}$: the accepted tensions, plus **Containment**: one cannot assert and deny the same sentence
- The base is **substructural**: **nonmonotonic** (adding premises can defeat an inference) and **nontransitive** (chains of good inferences need not compose): what defeasible, context-sensitive expert knowledge needs
- The NMMS sequent calculus elaborates logical vocabulary over the base: **supraclassical**, **conservative**, **explicitating**. It lets us *say* that a material inference holds without flattening it into a classical conditional
- **Traceability**: for every material implication, the transcript records when the tension was raised, what prompted it and how it was resolved: a justification that knowledge bases extracted from corpora lack

# Case study: Section 3.1 of the W3C PROV-O Recommendation

- **Input**: Section 3.1, "Starting Point Terms" (Lebo et al., 2013): 350 words of prose defining `Entity`, `Activity`, `Agent` and ten properties; examples, Turtle and figures excluded
- **Respondent**: a principal member of the W3C Provenance Incubator Group and co-editor of the PROV specifications. **Opponent**: Claude Opus 4.6. Two sessions, February 2026; the whole record is public
- **Result**: 10 initial commitments → 7 challenges → 19 commitments, **9 material implications**, one retraction, no open tensions
- **Ground truth**: Moreau et al. (2015) reconstructed PROV's design rationale *post hoc* from the working group's record of **8,820 emails, 666 issues and 152 teleconferences**: 73 requirements in nine themes
- **Six of the seven challenges correspond directly to design tensions documented in that reconstruction**; the requirements not surfaced lie almost entirely outside Section 3.1
- Rationale that is today reconstructed post hoc from mailing lists and issue trackers is captured **prospectively, as a byproduct of the design dialogue itself**

# PROV-O dialectic: atomic propositions

| Issue # | Commitment | Proposition |
|---|---|---|
| 1 | Three core classes form the basis of PROV-O: Entity, Activity, Agent | $p_1$ |
| 2 | Entity is a thing with fixed aspects | $p_2$ |
| 3 | Activity is something that occurs over time and acts upon or with entities | $p_3$ |
| 4 | Agent bears responsibility for activities, entities, or other agents' activities | $p_4$ |
| 5 | `used` and `wasGeneratedBy` relate Activities to Entities | $p_5$ |
| 6 | `wasInformedBy` provides Activity-to-Activity dependency | $p_6$ |
| 7 | `wasDerivedFrom` expresses Entity-to-Entity transformation | $p_7$ |
| 8 | `wasAssociatedWith` and `wasAttributedTo` ascribe Agent responsibility | $p_8$ |
| 9 | `actedOnBehalfOf` expresses delegation with shared responsibility | $p_9$ |
| 10 | Three types of provenance chains: Activity-Entity, Activity-only, Entity-only | $p_{10}$ |

# PROV-O dialectic: material implications

| Issue # | Challenge | (Moreau et al., 2015) | Resolution | Proposition | Material Implication |
|---|---|---|---|---|---|
| 11 | Fixed aspects; entity change? | RE1, RE3, RE5, RE6, EZ1 | Pragmatist: context-relative individuation | $p_{18}$ | $p_2 \mid\sim p_{18}$ |
| 12 | Activity duration; instants? | EV1, EV4 | Durational activities; instantaneous events | $p_{27}$ | $p_3 \mid\sim p_{27}$ |
| 13 | Responsibility vs. causation? | XG5, VI2–4, GE1 | Agency is pragmatic ascription | $p_{29}$ | $p_4 \mid\sim p_{29}$ |
| 14 | `wasInformedBy`: shortcut? | VI1, VI6, EZ1 | Independent: inferred, not reducible | $p_{30}$ | $p_6 \mid\sim p_{30}$ |
| 15 | Derivation criteria; identity? | XG8, VI5, VI6 | Broad causal dependencies; subtypes narrow | $p_{28}$ | $p_7 \mid\sim p_{28}$ |
| 16 | Delegation responsibility? | XG11 (partial) | Hierarchical and transitive | $p_{25}, p_{26}$ | $p_9 \mid\sim p_{25}, p_9 \mid\sim p_{26}$ |
| 17 | Chain types equivalent? | VI1, VI6 | No; `wasDerivedFrom` requires assertion | $p_{24}$ | $p_{10} \mid\sim p_{24}$ |
| — | Individuation $\Rightarrow$ Expanded | (from #11 follow-up) | Expanded Terms add expressiveness | $p_{23}$ | $p_{18} \mid\sim p_{23}$ |

# Verified with an automated reasoner: structure = design rationale

- pyNMMS, an open-source reasoner for the NMMS sequent calculus, checks the base: all 9 implications derivable, Containment holds, and all 34 cross-chain pairs are underivable, so each challenge addressed a distinct design tension
- Each structural property of the base corresponds to a documented design decision:

| Property | Queries (result) | Documented rationale (Moreau et al., 2015) |
|---|---|---|
| Nontransitivity | $p_2 \mid\!\sim p_{18}$ (T), $p_{18} \mid\!\sim p_{23}$ (T), $p_2 \mid\!\sim p_{23}$ (F) | EZ3: core/extended boundary |
| Derivation non-transitivity | $p_7 \mid\!\sim p_{28}$ (T); composed with other chains (F) | VI5, ISSUE-612: derivation not mandated transitive |
| Three-views independence | 8 cross-view queries, e.g. $p_2 \mid\!\sim p_{27}$, $p_4 \mid\!\sim p_{18}$ (all F) | VI1, GE1: no inference leaks between views |
| Nonmonotonicity | $p_2 \mid\!\sim p_{18}$ (T), $p_2, p_{23} \mid\!\sim p_{18}$ (F) | EZ1: "scruffy" to "proper" provenance |
| Retraction leaves no trace | $p_7 \mid\!\sim p_{18}$ (F), $p_7 \mid\!\sim p_{23}$ (F), $p_{18} \mid\!\sim p_{23}$ (T) | GE3: retracted $p_{20}$ left no residue |
| Explicitation | $\mid\!\sim p_2 \to p_{18}$ (T, all 9), $\mid\!\sim p_2 \to p_{23}$ (F) | EZ3: the chain is stated, not detached |

# Published as RDF, with its provenance in PROV-O

```
:i1 a mb:MaterialImplication, prov:Entity ;
    rdfs:label "p2 |~ p18" ;
    mb:antecedent :p2 ; mb:consequent :p18 ;
    prov:wasGeneratedBy :move11 .
:move11 a mb:DialogueMove, prov:Activity ;
    rdfs:label "Challenge #11 resolved by refinement" ;
    prov:wasAssociatedWith :respondent, :opponent ;
    prov:endedAtTime
        "2026-02-09T00:00:00Z"^^xsd:dateTime .
```

One material implication and the dialogue move that generated it, from `material-base.ttl`

- Propositions are resources; each material implication is reified and `prov:wasGeneratedBy` the dialogue move that accepted it, which links to the GitHub issues
- Validates against PROV-CONSTRAINTS (ProvValidator); the report is archived with the data
- The vocabulary the case study examined describes the dialectic that examined it

# Aligning with the RDF/OWL stack

- The base serializes to RDF, but its consequence relation **cannot be delegated to RDFS/OWL entailment without loss**: monotonic, transitive entailment would license exactly the sequents the base rejects. Resolved since the paper:
- **An implication-space semantics for RDF** (Allen, 2026b), after Hlobil (2026): simple, RDFS and rule-defined OWL 2 RL entailment, blank nodes and inconsistency included, recovered *exactly* as the **monotone corner** of the base's semantics, by the same homophonic reading: each triple's content is its own bearer's role. Lean-verified; matches `owlrl` on ∼8M graph pairs
- So regime closure and *defeasible* commitments (denials, defaults with exceptions, curator overrides) sit in **one consequence relation**, with no change to RDF, RDFS, OWL 2 RL or SHACL: a step toward the "semantics for shared inference" of Hitzler et al. (2010)
- **In practice**: pyNMMS over Oxigraph runs NMMS proof search against the regime's closure at $10^7 - 10^8$ triples, cost flat in graph size. Gene Ontology: monotone RDFS commits 394 gene products to classes their curators denied; as defaults defeated by the curators' `NOT`, none, and the 335 contradictions in the data surface for a curator to resolve

# Where this sits

## LLM generates, expert validates

- OntoChat: user stories, CQ extraction, filtration, testing (Zhang et al., 2025); participatory prompting (Zhao et al., 2024)
- RevOnt: CQs from existing knowledge graphs (Ciroku et al., 2024); NeOn with LLM automation (Kampars et al., 2025)
- LLMs as a *source* of knowledge: plausible but incorrect formalizations

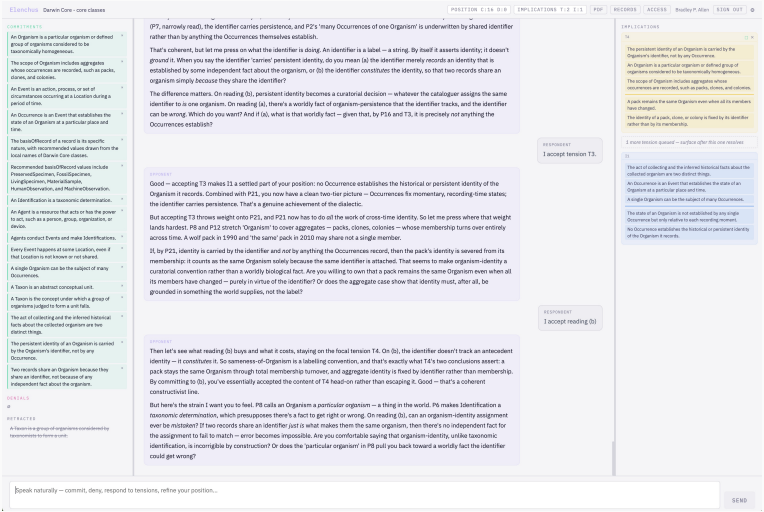
## Expert asserts, LLM challenges

- The LLM is a dialectical partner, not a source: hallucination becomes a contested, recorded challenge, and the respondent is the epistemic authority
- CQs and OWL fragments: no characterized consequence relation, no proven structural properties, no traceability. A material base has all three
- No knowledge engineer mediates between expert and formalism (cf. CommonKADS, METHONTOLOGY, NeOn)

# Evaluation: what the case study does and does not show

- The case study is an **existence proof**, not an evaluation: the respondent co-authored the retrospective analysis used as ground truth, and the opponent's training data almost certainly includes the working group's public record, so its challenges may be recall rather than detection. Neither affects the protocol's guarantees, but both limit what the case study shows
- **A funded, pre-registered study** (Sloan Foundation G-2026-79650): thirty biodiversity scientists each specify a matched pair of domains under two counterbalanced conditions, free-form LLM conversation and Elenchus, model held fixed; a blinded panel of five expert judges rates the outputs on established ontology-quality criteria; one pre-registered confirmatory test
- **Infrastructure**: elenchus-server, a second, independent implementation (model-agnostic, multi-user, typed speech-act API, derivability queries, DuckDB), and pyNMMS over RDF/Oxigraph
- **Known limitations**: systematic oracle bias (false negatives are invisible to both parties); scaling beyond a single specification section; the protocol is bilateral, so reconciling divergent positions across respondents is open

# elenchus-server user interface



# In closing

- The intended setting is **ontology working groups and enterprise vocabulary governance**, where the deliverable is not only the ontology but **a defensible record of why it is the way it is**
- The expert is never asked to translate their knowledge into a target formalism: the dialogue elicits the consequence relation itself
- The LLM's unreliability is contained by the protocol, not managed after the fact
- Everything is public:
  - Protocol and prompt: `https://github.com/bradleypallen/elenchus`
  - PROV-O dialectic, material base and export: `https://github.com/bradleypallen/prov-o-section-3.1`
  - Reasoner: `https://pypi.org/project/pyNMMS/`
  - Server: `https://pypi.org/project/elenchus/`

# References i


Allen, Bradley P. (2025). **"Conceptual Engineering Using Large Language Models".** In: *Philosophy of Artificial Intelligence: The State of the Art)*. Ed. by Vincent C. Müller et al. To appear. Springer Nature. url: https://doi.org/10.48550/arXiv.2404.03732.

— (2026a). ***Elenchus: Generating Knowledge Bases from Prover-Skeptic Dialogues.*** arXiv:2603.06974. url: https://arxiv.org/abs/2603.06974.

— (2026b). **"Implication-Space Semantics for RDF".** Preprint; arXiv number pending. Lean 4 formalization and machine checks at https://github.com/bradleypallen/iss-rdf-lean and https://github.com/bradleypallen/iss-rdf-checks.

Allen, Bradley P. and Paul Groth (2025a). **"Evaluating Class Membership Relations in Knowledge Graphs using Large Language Models".** In: *The Semantic Web: ESWC 2024 Satellite Events*. Ed. by Albert Meroño-Peñuela et al. Vol. 15344. Lecture Notes in Computer Science. Cham: Springer. isbn: 978-3-031-78951-9. doi: 10.1007/978-3-031-78952-6_2. url: https://doi.org/10.1007/978-3-031-78952-6_2.

Allen, Bradley P. and Paul T. Groth (2025b). **"A Benchmark for the Detection of Metalinguistic Disagreements between LLMs and Knowledge Graphs".** In: *Proceedings of the Special Session on Harmonising Generative AI and Semantic Web Technologies (HGAIS 2024) co-located with the 23rd International Semantic Web Conference (ISWC 2024)*. Vol. 3953. CEUR Workshop Proceedings. Baltimore, Maryland: CEUR-WS.org. url: https://ceur-ws.org/Vol-3953/.

Allen, Bradley P., Lise Stork, and Paul Groth (2023). **"Knowledge Engineering Using Large Language Models".** In: *Transactions on Graph Data and Knowledge* 1.1, 3:1–3:19. doi: 10.4230/TGDK.1.1.3.

Allen, Bradley P. et al. (2025c). **"Sound and Complete Neurosymbolic Reasoning with LLM-Grounded Interpretations".** In: *Proceedings of The 19th International Conference on Neurosymbolic Learning and Reasoning*. Ed. by Leilani H. Gilpin et al. Vol. 284. Proceedings of Machine Learning Research. PMLR, pp. 392–419. url: https://proceedings.mlr.press/v284/allen25a.html.

## References iii


Augier, Mie-Sophia and Morten Thanning Vendelø (2002). ***An Interview with Edward A. Feigenbaum.*** Interview. Copenhagen, Denmark: Copenhagen Business School, Department of Informatics. url: https://www.researchgate.net/publication/236904576_An_Interview_with_Edward_A_Feigenbaum.

Brandom, Robert (1994). ***Making it explicit: Reasoning, representing, and discursive commitment.*** Harvard University Press.

Ciroku, Fiorela et al. (2024). **"RevOnt: Reverse engineering of competency questions from knowledge graphs via language models".** In: *Journal of Web Semantics* 82, p. 100822.

Dutilh Novaes, Catarina (2012). ***Formal languages in logic: A philosophical and cognitive analysis.*** Cambridge University Press.

— (2020). ***The dialogical roots of deduction: Historical, cognitive, and philosophical perspectives on reasoning.*** Cambridge University Press. doi: 10.1017/9781108800792.

Forsythe, Diana E (1993). **"Engineering knowledge: The construction of knowledge in artificial intelligence".** In: *Social studies of science* 23.3, pp. 445–477.

Garijo, Daniel et al. (2024). **"LLMs for ontology engineering: a landscape of tasks and benchmarking challenges".** In: *Proceedings of the Special Session on Harmonising Generative AI and Semantic Web Technologies (HGAIS 2024) co-located with the 23rd International Semantic Web Conference (ISWC 2024)*. Vol. 3953. CEUR Workshop Proceedings. CEUR-WS.org, pp. 1–7. url: https://ceur-ws.org/Vol-3953/364.pdf.

Hitzler, Pascal and Frank van Harmelen (2010). **"A reasonable Semantic Web".** In: *Semantic Web* 1.1–2, pp. 39–44. doi: 10.3233/SW-2010-0010.

Hlobil, Ulf (2026). **"First-order implication-space semantics".** In: *Journal of Philosophical Logic* 55.3, pp. 529–554. doi: 10.1007/s10992-026-09841-x.

Hlobil, Ulf and Robert B Brandom (2025). ***Reasons for logic, logic for reasons: Pragmatics, semantics, and conceptual roles*.** Routledge.

Huang, Jiacheng et al. (2026). **"Socratic Elenchus-inspired multi-agent debate for mitigating hallucinations in large language models".** In: *Expert Systems with Applications* 320, p. 132208. doi: 10.1016/j.eswa.2026.132208.

Kampars, Janis et al. (2025). **"LLM-supported collaborative ontology design for data and knowledge management platforms".** In: *Frontiers in Big Data* 8, p. 1676477.

Khan, Akbir et al. (2024). **"Debating with More Persuasive LLMs Leads to More Truthful Answers".** In: *Proceedings of the 41st International Conference on Machine Learning*. Vol. 235. Proceedings of Machine Learning Research. PMLR, pp. 23662–23733. url: https://proceedings.mlr.press/v235/khan24a.html.

Koutsiana, Elisavet et al. (2026). **"Knowledge prompting: How knowledge engineers use generative AI".** In: *Journal of Web Semantics* 88, p. 100873. doi: 10.1016/j.websem.2025.100873.

Lebo, Timothy, Satya Sahoo, and Deborah McGuinness (2013). ***PROV-O: The PROV Ontology*.** W3C Recommendation. W3C. url: https://www.w3.org/TR/prov-o/.

Liang, Tian et al. (2024). **"Encouraging divergent thinking in large language models through multi-agent debate".** In: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 17889–17904.

Moreau, Luc et al. (2015). **“The rationale of PROV”.** In: *Journal of Web Semantics* 35, pp. 235–257.

Restall, Greg (2005). **“Multiple conclusions”.** In: *Logic, methodology and philosophy of science: Proceedings of the twelfth international congress*. Kings College Publications, pp. 189–205.

Sellars, Wilfrid (1953). **“Inference and meaning”.** In: *Mind* 62.247, pp. 313–338.

Studer, Rudi, V Richard Benjamins, and Dieter Fensel (1998). **“Knowledge engineering: Principles and methods”.** In: *Data & knowledge engineering* 25.1-2, pp. 161–197. doi: 10.1016/S0169-023X(97)00056-6.

Zhang, Bohui et al. (2025). **“OntoChat: A Framework for Conversational Ontology Engineering Using Language Models”.** In: *The Semantic Web: ESWC 2024 Satellite Events*. Vol. 15344. Lecture Notes in Computer Science. Cham: Springer, pp. 102–121. doi: 10.1007/978-3-031-78952-6_10.

Zhao, Yihang et al. (2024). **“Improving ontology requirements engineering with OntoChat and participatory prompting”.** In: *Proceedings of the AAAI Symposium Series*. Vol. 4:1, pp. 253–257.